

New Relic Monitoring – Documentation

Overview

New Relic is integrated to provide:

- **Application Performance Monitoring (APM)** for the Java Spring Boot API
- **Infrastructure & container metrics** for ECS Fargate
- **Log forwarding** to New Relic Logs
- **Alerting** based on application and infrastructure health

Monitoring is enabled **per environment** (dev / prod) and follows **least-privilege and secret-based authentication**.

Architecture Integration

Components Involved

Component	Purpose
New Relic Java Agent	APM instrumentation
FireLens (Fluent Bit)	Log forwarding
New Relic Infrastructure Agent	Container & ECS metrics
AWS Secrets Manager	Secure license key storage
ECS Task Definition	Agent and log configuration

How New Relic Is Integrated

APM (Java Application)

- The **New Relic Java Agent** is bundled into the application Docker image.
- The agent automatically instruments:
 - HTTP requests
 - JVM metrics
 - Garbage collection
 - Database calls (if applicable)

Environment Variables (Injected into ECS Task)

```
NEW_RELIC_APP_NAME=java-api-<environment>  
NEW_RELIC_DISTRIBUTED_TRACING_ENABLED=true
```

Infrastructure & ECS Metrics

- A **sidecar container** runs the New Relic Infrastructure agent.
- The agent detects:
 - ECS cluster
 - Service
 - Task metadata
 - CPU, memory, and task health

```
Container: newrelic/infrastructure:latest
Mode: Fargate-compatible
```

Log Forwarding

- Logs are routed via **FireLens (Fluent Bit)**.
 - Logs are sent directly to New Relic's log ingestion endpoint.
 - No logs are publicly exposed.
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Required Configuration

Secrets & Credentials

Item	Storage	Used By
New Relic License Key	AWS Secrets Manager	ECS Task
IAM permissions	IAM Role	ECS Task

AWS Secrets Manager

Secret name: newrelic/license-key

Injected into ECS using:

```
secrets = [{
  name      = "NEW_RELIC_LICENSE_KEY"
  valueFrom = var.newrelic_secret_arn
}]
```

Security

- License key is never stored in code or Terraform state
 - Access is restricted to ECS Task Role only
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IAM Permissions (Least Privilege)

ECS task role requires:

- `secretsmanager:GetSecretValue`
 - `logs:PutLogEvents`
 - `logs:CreateLogStream`
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Viewing Metrics in New Relic

Access Steps

1. Log in to **New Relic One**
 2. Navigate to:
 - **APM & Services** → **java-api-<environment>**
 - **Infrastructure** → **ECS**
 - **Logs** → **Filter by service name**
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Key Dashboards

Application (APM)

- Response time
- Throughput (RPM)
- Error rate
- JVM heap usage
- GC pauses

Infrastructure

- ECS service CPU & memory
- Task restarts
- Container health
- Deployment markers

Logs

- Application logs
 - Error stack traces
 - Startup & shutdown events
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Alert Configuration

Alert Policy Structure

Alerts are configured using **New Relic Alert Policies**.

Recommended Alerts

Application Alerts

Metric	Threshold
Error rate	> 5% for 5 minutes
Response time	> 1s for 5 minutes
Apdex score	< 0.85

Infrastructure Alerts

Metric	Threshold
CPU Utilization	> 80% for 10 minutes
Memory Utilization	> 75% for 10 minutes
Task restarts	> 2 in 5 minutes

Availability Alert

- **No data received** for 5 minutes
 - Detects crashed or stuck services
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Alert Destinations

- Email
 - Slack
 - PagerDuty (optional)
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Deployment Markers

Each deployment creates a **deployment marker** in New Relic:

- Correlates performance changes with releases
- Helps identify regressions

Triggered during CI/CD deployment to `main`.

Environment-Based Enablement

Monitoring can be toggled:

```
enable_newrelic = true
```

Environment Enabled

dev	Optional
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prod	Required
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